Export Control Form for GLOBALFOUNDRIES

Customer	Elijah Tate Johnson	**Device Name**	Four-Topology MOSbius
Project Name	Four-Topology MOSbius	**Phone**	+1 (865) 705-1811
Contact Person	Elijah Tate Johnson	**Email ID**	etj2114@columbia.edu

1. a) Please provide a **brief description** of the product (i.e., chiplet, structure, etc) to be manufactured by GLOBALFOUNDRIES and the final end use for the product (for example, "baseband for mobile phone" or "microcontroller for industrial equipment" or "application processor for media player" or "automotive system-on-chip").	Analog circuit design intuition is best developed through hands-on measurement of real silicon, yet most educational settings rely primarily on simulation. Four-Topology MOSbius is an open-source, programmable amplifier test chip that enables students to configure and evaluate four distinct amplifier topologies, a folded cascode OTA, a five- transistor OTA, a telescopic cascode OTA, and a common- source stage, on a single die. Each amplifier is implemented with programmable transistors that allow selection among four sizing configurations (base through 4× sizing), and both topology and sizing are selected at runtime through an on-chip scan chain without powering down the system. All amplifiers were designed using the gm/ID methodology to target a 1 MHz gain- bandwidth product while driving a 120 pF off-chip load, and the design was completed entirely with open-source tools (Xschem, Ngspice, and KLayout) and the GlobalFoundries GF180MCU open-source PDK. Building on the MOSbius platform, the chip supports side-by-side comparison of gain, bandwidth, slew rate, phase margin, and power consumption under identical loading conditions, as well as two-stage configurations with external frequency compensation

1. b) **(i)** Provide the peak saturated power output	N/A
(ii) Provide the operating frequency	N/A
(iii) Provide the fractional bandwidth	N/A
(iv) Provide the planar side length	N/A
1. c) Is the product a millimeter wave (mmWave) product, such as monolithic microwave integrated circuit (MMIC) amplifier or discrete microwave transistor?	No
(i) Attach a graph showing the x-axis (frequency) and y-axis (peak saturated power output) and as applicable a graph (i.e., S21 plot) with x-axis (frequency) and y-axis (power gain)	
(ii) Does this product include an integrated phase shifter?	N/A
(iii) Is this product "specially designed" for other applications e.g., telecommunications, radar, automobiles?	N/A
If yes, please specify:	N/A
(iv) Is the product or related design controlled under any of the following: U.S. Munitions List XI(c)(4) for military transmit/receive MMICs, XV(e)(14) for space-qualified MMICS; or US ECCNs 3A001.b.2 for general purpose MMICs, or 3A001.b.12 transmit/receive MMICS, or 5E001.d for telecommunications MMIC technology?	N/A
If yes, please specify:	N/A

2. Is any party to your transaction involving the products identified on a trade controls restriction list, 50% or more owned by such a party, or acting on behalf of or for the ultimate benefit of any such party? Trade controls restriction lists include, without limitation, the U.S. Entity List, U.S. List of Denied Persons, U.S. Unverified List, U.S. List of Military End Users, the U.S. Specially Designated Nationals and Blocked Person List, the U.S. Sectoral Sanctions List, or any other list maintained by the United States or other applicable jurisdiction, the effect of which is to restrict exports, imports, payments, services, or other transactions with such parties?	No
3. Is any party to your transaction organized under the laws of, located in, or ordinarily resident in a country or region subject to comprehensive or substantial trade restrictions, presently including Belarus, Cuba, Iran, North Korea, Russia, Syria, or the occupied regions of Ukraine (presently Crimea/Sevastopol, Donetsk, Kherson, Luhansk, Zaporizhzhia)?	No
4. Will the tapeout for this project include more than one chip design?	No
5. Are the product and its related design/technology or software subject to export controls in non-U.S. Jurisdictions? If YES, please state the jurisdiction and applicable controls	No
6. Will this chiplet include variants that will not change the "end use" and/or "export control" outcome?	No
7. Is the product an application specific integrated circuit (ASIC) or otherwise unalterably programmed or designed for a specific function for other equipment? If yes, Please describe the specific application, function, or equipment	No

8. Is the product intended or likely to be used for or in connection with the development, production, handling, operation, maintenance, storage, detection, identification or dissemination of any nuclear, chemical or biological weapon; unsafeguarded nuclear activities, nuclear materials production, or maritime nuclear propulsion systems; or the development, production, maintenance or storage of missiles, rocket systems, or unmanned aerial vehicles capable of a range of at least 300 kilometers or otherwise intended for delivering weapons?	No
9. U.S. Entity List supply chain restrictions: Does your company have knowledge that the product to be delivered by GF will be exported outside the United States, re-exported, or in-country transferred abroad for any of the following? i. incorporation into or use in the production or development of any product to be produced, purchased or ordered by any party on the U.S. Entity List(https://www.ecfr.gov/current/title-15/subtitle-B/chapter-VII/subchapter-C/part-744/appendix-Supplement%20No.%204%20to%20Part%20744) that is designated with Footnotes 1, 3, 4 or 5; or ii. a transaction to which any party on the U.S. Entity List that is designated with Footnotes 1, 3, 4 or 5(https://www.ecfr.gov/current/title-15/subtitle-B/chapter-VII/subchapter-C/part-734/section-734.9#p-734.9(e)) is a party, such as a purchaser, intermediate/ultimate consignee, or end-user? The Export Administration Regulations define "Knowledge" in section 772.1 as [https://www.ecfr.gov/current/title-15/subtitle-B/chapter-VII/subchapter-C/part-772/section-772.1].	No

10. Are the product (i.e., wafer, die, module), the associated technology/technical data (i.e., integrated circuit design data), and any software specially designed, modified, or adapted for a military or intelligence application, or intended for incorporation or end use in an article subject to ITAR, EAR 500- or 600-Series, or another jurisdiction's Munitions controls? If yes, please answer the questions below: **I.** Where available, please provide supporting documents for the classification and jurisdictional determination e.g., US State Department Commodity Jurisdiction (CJ) determination, U.S. Commerce Department Classification Ruling(CCATS), advisory opinion or ruling from other agency or jurisdiction, or self classification memo prepared internally or by outside counsel or trade consultants. Do you have such supporting documents? **i.** If known, describe how the product has been specially designed, modified, or adapted for military or intelligence application **ii.** If known, describe the intended end use of the product and article into which it is being incorporated: **iii.** Provide the full ITAR USML or EAR 500/600-series classification at the sub paragraph levels that apply to the product (i.e., wafer, die, module). When applicable ,provide the non -US Munitions control. **iv**. Provide the full ITAR USML or EAR 500/600-series classification at the sub paragraph levels that apply to the Design file (i.e., GDSII)/ Technology. When applicable, provide the non-US Munitions control. **v.** If known, provide the full ITAR USML or EAR 500/600-series classification at the sub paragraph levels, and any applicable non US Munitions control, that apply to the next higher assembly of the Product that GF is fabricating.	No

11. Is the product (i.e., wafer, die, module), associated technical data (i.e., integrated circuit design data), and any software specially designed or modified to use encryption? If yes, please answer the questions below: **i.** Please describe the applicable encryption protocols, algorithms key lengths, and the source of the encryption. **ii:** Was the product, technical data, and software reviewed by the US Commerce Department pursuant to a CCATS? If yes, provide the CCATS# under(iii)and the CCATS Date under(v). If no, indicate N/A under (iii) and (v). **iii:** Was the product, technical data, and software self-classified? If yes, provide the self-classification date in the free text box below and if applicable, under field (v) provide the date on which a self-classification report was filed with the U.S. Commerce Department **iv**: Provide CCATS number. **v**: Where available, please provide supporting documents for the classification and jurisdictional determination, e.g., U.S. State Department Commodity Jurisdiction (CJ) determination, U.S. Commerce Department Classification Ruling (CCATS), advisory opinion or ruling from other agency or jurisdiction, or self-classification memo prepared internally or by outside counsel or trade consultants. **vi:** Provide CCATS Date (or) Self-Classification Report Date to U.S. Commerce Department **vii**: If the product is subject to U.S. encryption controls under the EAR, state the ENC § 740.17 sub paragraph for the product, technical data, and software: (b)(2)(i)(A) Network infrastructure (b)(2)(i)(B) Encryption source code	No

12. Is the product (i.e. wafer, die, module) or the associated technical data (i.e. integrated circuit design data) subject to the U.S. Export Administration Regulations ("EAR", 15 CFR Part 730 et seq.)? If yes, please answer below questions: **i.** Where available, please provide supporting documents for the classification and jurisdictional determination, e.g., U.S. State Department Commodity Jurisdiction (CJ) determination, U.S. Commerce Department Classification Ruling (CCATS), advisory opinion or ruling from other agency or jurisdiction, or self-classification memo prepared internally or by outside counsel or trade consultants **ii.**Provide the U.S. ECCN at the sub-paragraph level. **Product (i.e., wafer, die, module), Design File/Technology, Software (if any, otherwise apply N/A)**	No

13. After review of the 3A090 ECCN Notes shown, and regardless of how the product is ultimately controlled. please provide a response to the following question. Does the item also meet or exceed the performance parameters of ECCN 3A090 on the U.S CCL ? Notes – ECCN 3A090: a. Integrated circuits having one or more digital processing units having either of the following: a.1. A total processing performance of 4800 or more, or a.2. A total processing performance of 1600 or more and a 'performance density' of 5.92 or more. b. Integrated circuits having one or more digital processing units having either of the following: b.1. A total processing performance of 2400 or more and less than 4800 and a 'performance density' of 1.6 or more and less than 5.92, or b.2. A total processing performance of 1600 or more and a performance density of 3.2 or more and less than 5.92. c.High bandwidth memory (HBM) having a 'memory bandwidth density' greater than 2 gigabytes per second per square millimeter. If yes, please answer the question below:	No
i. Is your company an "Approved IC Designer" listed in the U.S. Export Administration Regulations ("EAR"), Supplement No. 6 to 15 C.F.R Part 740? Supplement No. 6 to Part 740 - Approved integrated Circuit Designers | Bureau of Industry and Security(https://portal.myglobalfoundries.com/tapeout/export-control/edit/chiplet/?id=4afa7d86-d58f-f011-b4cc-000d3a1395ca&hslexportcontrolid=90dd267e-858f-f011-b4cc-000d3a5650ac&mode=edit&stepid=26a8c87a-4758-ee11-be6f-000d3a18283a&sessionid=c4b22111-da8f-f011-b4cc-000d3a57b6d3). If answer is No, please answer to the below question:	No
i. Does your company meet the criteria for "Authorized IC Designer" per the EAR : • Headquartered in Taiwan or a country in EAR Country Group A:1 or A:5; • Not located in Macau or any Country in EAR Country Group D:5; • Does not have an ultimate parent company headquartered in Macau or any Country in EAR Country Group D:5; • Has submitted, or is in the process of submitting, information described in EAR Section 743.9(c) to GlobalFoundries, the front-end fabricator, to enable its quarterly reporting.	No

14. Does the product meet any of the technical criteria in any other ECCN on the CCL? If yes, please state which ECCN.	No
15. Is the product a qubit device or qubit circuit, containing or supporting arrays of physical qubits and specially designed for quantum computers? If yes please answer to below questions: **i.** Provide the number of fully controlled, connected and working physical qubits supported by the related quantum computer ii. Provide the C-NOT error	No
16. If you have answered 'no' to questions 10,11 and 12,that indicates that the Product (i.e. wafer ,die ,module) and its associated technology (i.e. integrated circuit design data) are subject to export controls in (non-US) jurisdictions. Please answer YES below and provide the non-US export control classification for each field. If not subject to (non-US) jurisdictions, please answer NO and revisit the former questions before proceeding. Please also be aware of FDPR(Foreign Direct Product Rule) (https://www.ecfr.gov/current/title-15/subtitle-B/chapter-VII/subchapter-C/part-734/section-734.9#p-734.9%28e%29) definition below and revisit questions 10,11 and 12 if needed. Foreign Direct Product Rule(https://www.ecfr.gov/current/title-15/subtitle-B/chapter-VII/subchapter-C/part-734/section-734.9#p-734.9%28e%29): You must assume that any item produced by GF, at any of its fabs no matter where located, is subject to U.S. Foreign Direct Product Rule on the basis that GF's production process and plant equipment invoke the product scope in 15 C.F.R. § 734.9 (e), (f), (g), (h), (i), (j) and (k). **If yes, please provider Product (i.e., wafer, die, module), Design File/ Technology, Software (if any, otherwise apply N/A)**	Yes Wafer

17. Please provide the Harmonized System classification for the finished wafers. Under Heading 85.42 and subcategories of Electronic Integrated Circuits.

Select One	HS Code	Description
Y	**8542.31**	Processor and controllers, whether or not combined with memories, converter, logic circuits, amplifier, clock and timing circuits, or other circuits.
☐	8542.31.0015	8-bit No.
☐	8542.31.0020	16-bit No.
☐	8542.31.0025	32-bit No.
☐	8542.31.0030	Other No. –
☐	8542.31.0035	Digital signal processors (DSPs) No.
☐	8542.31.0040	Graphics processing units (GPUs) No.
☐	8542.31.0045	Central processing units (CPUs) No.
☐	8542.31.0050	Other No. –
☐	8542.31.0055	Complex programmable logic devices (CPLDs) No.
☐	8542.31.0060	Field programmable gate arrays (FPGAs) No.
☐	8542.31.0065	Other No.
☐	8542.31.0070	System on chip (SoCs) No.
☐	8542.31.0075	Other No.
☐	**8542.32**	Memories
☐	8542.32.0015	Not over 1 gigabit No.
☐	8542.32.0023	Over 1 gigabit No.
☐	8542.32.0040	Static read-write random access (SRAM) No.
☐	8542.32.0050	Electrically erasable programmable read-only memory (EEPROM)
☐	8542.32.0060	Erasable (except electrically) programmable read-only memory (EPROM) No.
☐	8542.32.0070	Other No.

18a. Is the product intended or likely to be used for or in connection with any military end use/end user or military-intelligence end use/user connected with Belarus, Burma (Myanmar), Cambodia, China, Iraq, Nicaragua, Russia, or Venezuela? "Military end use" means 744.21(f) [here]. "Military end user" means 744.21(g) [here]. "Military-intelligence end user" means in 744.22(f) [here] If Yes, answer to below question: **i.**Is the product to be delivered by GLOBALFOUNDRIES intended for a military end use or military end user and classified under one of the following ECCNs: 3A991, 3A992, 5A991.a, 5A991.b.7, 5A991.f, 5A992, or is the product to be delivered by GLOBALFOUNDRIES intended for a military end-use or a military end-user in the Russian Federation or Belarus, regardless of export classification (including classification under EAR99)?	No
18b. If the product is a microprocessor (i.e., microprocessor microcircuits, microcomputer microcircuits, and microcontroller microcircuits having a processing speed of 5 GFLOPS or more and an arithmetic logic unit with an access width of 32 bit or more), including those incorporating information security functionality, or associated software and technology for the production or development of such microprocessors, will it be intended or likely used for or in connection with any military end use/end user in a destination listed in EAR Country Group D:1, presently including: Armenia, Azerbaijan, Belarus, Burma, Cambodia, China (PRC), Georgia, Iraq, Kazakhstan, Korea (North), Kyrgyzstan, Laos, Libya, Macau, Moldova, Mongolia, Nicaragua, Russia, Tajikistan, Turkmenistan, Uzbekistan, Venezuela, Vietnam, Yemen	No

19. Advanced chip and computing end use restrictions: Does your company have knowledge that the product to be delivered by GF will be exported outside the United States, re-exported, or in-country transferred abroad for any of the following? i).the items will be used in the development or production of integrated circuits destined to a facility located in EAR Country Group D:5 or Macau, where production of advanced-node integrated circuits occurs; or ii).the items are specified in an ECCN in Product Groups B, C, D, or E in Category 3 of the EAR CCL, and will be used in the development or production of integrated circuits destined to a facility located in EAR Country Group D:5 or Macau, where you do not know if production of advanced-node integrated circuits occurs; or iii).items in ECCNs 3A090, 4A090, and .z items on the EAR CCL, and their related development and production technology, destined to EAR Country Groups D:1, D:4 or D:5, or to entities headquartered in or whose ultimate parent company is headquartered in Country Group D:5 or Macau; or iv).the items will be used in the design, development, production, operation, installation (including on-site installation), maintenance (checking), repair, overhaul or refurbishing of a supercomputer, or incorporated into or used in the development or production of any part, component or equipment that will be used in a supercomputer located in or destined to or in, either Macau or a destination specified in Country Group D:5. The EAR defines "Knowledge" in section 772.1 as [here]. The EAR defines "supercomputer" in section 772.1 as [here]	No
20. Who is the **end user** of the manufactured IC? (Enter the **name of the company** and the **full postal address.** In case of multiple end users, add each user!)	Elijah Tate Johnson 70 Morningside Drive, etj2114 EC, NY, New York 10027

Your electronic signature below certifies that you are authorized to sign this certification on behalf of Elijah Tate Johnson (“Organization”) and that the representations in it are true and correct to the best of your knowledge.

I, Elijah Tate Johnson. , certify that the above is true and correct to the best of my knowledge. I certify that I am duly authorized to bind Elijah Tate Johnson (“Organization”).

Name:

Title:

Signature:

X 